A 0.9V 64Mb 6T SRAM cell with Read and Write assist schemes in 65nm LSTP technology

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***Abstract*—A low Vmin, 6T SRAM is designed in 65nm LSTP (low standby power) technology using read and write assist schemes. In this work, we reduced the Vmin of SRAM cell from 1.2V to 0.9V by using assist schemes. Static noise margin (SNM) is improved by using Compensated Wordline Underdrive (WLUD) scheme as read assist. Using this read assist, maximum WLUD of 261mV is achieved for SNM critical PVT i.e., FS/0.9V/125C. SNM achieved is 465mV ensuring that it qualifies required cell sigma robustness. The underdrive at other PVT corners is lesser therefore it limits the impact on speed and write margin (WM) at their respective critical PVT. Negative BL approach using capacitive coupling is used as write assist in which a small capacitor of 10fF is placed in the write driver to achieve required bitline undershoot. BL undershoot obtained is 166mV which improves the WM at critical PVT i.e., SF/0.9V/-40C and thereby write-ability. WM at this PVT is 286mV. By enabling lower voltage of operation upto 45% of dynamic power reduction is achieved in the design.**

***Index Terms*—Static Random Access memory(SRAM), read assist, WM, SNM, 65nm**

I. Introduction

Continued CMOS Scaling, in line with Moore's Law, over the past few decades has enabled very high level of integration on SoC. CMOS Scaling has also led to design of smaller devices which are not only faster, but also consume lower power. This has improved overall system performance. However, as devices reduce in size, manufacturing variations increase. Increase in manufacturing variations of transistors is usually modelled as variations in threshold voltage (Vth). These device variations impact the functionality of circuits and limit their performance and area scaling. SRAM cell is one of the most important scaling limiting circuit on any chip[1].

Therefore the designers face a optimisation challenge between power and performance due to higher integration over SoC. There are several methods to reduce the power consumption of the chip, out of which lowering the operational voltage is one of the most effective way as it not only reduces the dynamic power but also the static power of the chip. SRAMs limit the minimum operational voltage on any chip[2]. They also occupy a large part of the chip area therefore the static power consumption is also very critical. If the voltage of SRAM array can be reduced, it can lead to multifold benefits in overall reducing the power consumption of a chip but reducing the voltage drastically degrades the SNM and WM of the cell.

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Out of several logic circuitry, SRAM cells are affected most by the parametric variations as they consist of minimum-sized transistors. Parametric variations are mainly of two types, namely Global variations and Local variations[3]. These variations affect the Figures of Merit (FoMs) of the SRAM cells such as SNM, WM etc. and may cause read or write failures inside the cells. Hence it is essential to accurately model the yield of the SRAM cells which is the commonly used method to estimate robustness and reliability of the design in the presence of process-induced variations[4].

SRAM yield can be understood as the total amount of process variations across devices that still lead to a functional cell where functionality is defined across read, write and the hold operations[5]. These aspects are explained using 'cell sigma', which is defined as mean value divided by the standard deviation for any read or write metric of SRAM cell that has a Gaussian distribution[6]. There are several methods to estimate the yield of the cell, namely Monte Carlo Simulations[1], Design Of Experiments[1], and Importance Sampling[7], out of which the most common approach to compute yield is Monte Carlo Simulations. In [8], authors proposed a simple and effective technique to estimate probability of failures during read cycle with respect to SNM.

The required yield can be correlated with the cell sigma by considering the SRAM capacity and the pass probability required from a particular cell. If the average failure rate of a component is X, then Y is the pass probability (Y=1-X). Here X=D/M where D is defect density or accepted failure rate and M is number of components. In further analysis we assume that a single defect will lead to a die failure and if there is more than one defect then as many number of dies will fail. In other words, we ignore marginal improvement in yield that might happen due to more than one defect occuring

TABLE I
Gaussian sigma for different SRAM capacity provided 1 fail

capacity(Mb)	X	(Y=1-X)	Gaussian sigma
1	9.53×10^{-9}	0.99999999046	5.6
4	2.38×10^{-9}	0.99999999761	5.85
8	1.19×10^{-9}	0.9999999988	5.97
16	5.96×10^{-10}	0.99999999940	6.1
64	1.49×10^{-10}	0.99999999985	6.3

on a die. Table I shows the Gaussian sigma for different SRAM capacity provided there is only one fail per 100 dies. The rest of the paper is organized as follows: In Section II 6T SRAM cell design is discussed. Section III discusses the read assist scheme. In Section IV write assist scheme is explained using circuit and timing diagram. Simulation results for read and write assist schemes are presented in Section V and finally we conclude our work in Section VI.

TABLE II
SNM, ICELL AND WM FOR DIFFERENT PVT @ 0.9V(NO ASSIST APPLIED)

PVT	SNM		WM		Icell
	(mV)	Cell sigma	(mV)	Cell sigma	(μA)
FS/125C	302	5.30	310	9.06	9.69
SS/-40C	384	6.85	174	5.32	4.525
SF/-40C	420	7.96	148	4.59	5.139
TT/125C	348	6.39	260	7.70	8.142
TT/-40C	377	6.91	195	5.93	6.811
FF/125C	330	6	274	8.11	10.82
FF/-40C	370	6.82	214	6.48	9.84

II. 6T SRAM CELL DESIGN

A 6T SRAM cell is designed with a cell ratio of 1.57 in 65 nm. The width of PU, PD and PG transistors are 120nm, 220nm, and 140 nm respectively and the length is 90nm. The cell is simulated at 1.2V followed by supply voltage lowering and it is found that at 0.9V SNM of the SRAM cell fails to qualify the required cell sigma for the worst corner, i.e., FS/125C. The values for SNM and WM, along with their cell sigma at 0.9V, can be seen from Table II. The values of Icell is also presented in Table II. Cell sigma of Icell is not used because at low voltages Icell distribution is not Gaussian. Hence cell sigma of Icell cannot be considered across PVTs. Several read and write assist schemes have been studied and implemented so far[2][6][9-13].We have implemented the Compensated Wordline Under-drive approach[2] as the read scheme on our cell to improve the SNM and achieve a Vmin of 0.9V. To improve the WM, we have implemented Bitline undershoot[2][9][14] using Capacitive Coupling as the write assist which makes it qualify desired cell sigma for WM.

III. READ ASSIST CIRCUIT

The assist circuit implemented here enhances the SNM and cell sigma robustness. Simulation setup having Compensated WLUD[2] circuit along with 1000 SRAM cells in a row is illustrated in Fig. 1. Wordline Underdrive (WLUD) implies that the voltage of the wordline is lowered while finding SNM of the cell.

There are critical PVT corners for every cell parameter like speed, WM, SNM. At these critical corners, the value of the corresponding parameter is worst. SNM is worst at FS/125C, cell current is worst at SS/-40C, WM is worst at SF/-40C. Compensated WLUD approach utilizes this concept of critical PVT Corners to improve the SNM without degrading cell current. This scheme lowers the WL voltage levels, not

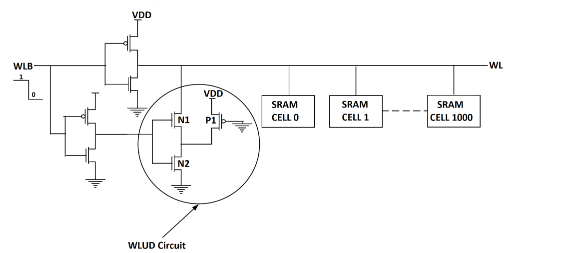

Fig. 1. Simulation Setup for Compensated WLUD as Read Assist Circuit

at all the corners but only at SNM critical PVT corner, i.e., FS/125C. This leads to improvement in SNM at FS/125C. At performance-critical PVT i.e., SS/-40C and WM critical PVT i.e., SF/-40C, the WLUD is very less which does not impact the WL voltage and hence cell current and WM remains as it is.

Compensated WLUD ensures that the appropriate lowering of the WL voltage occurs at SNM critical PVT only. WLUD circuit consists of a stack of two NMOS N1 and N2 along with a PMOS P1 whose gate is grounded. When *WLB* voltage is applied (1 to 0) to the upper CMOS inverter, a voltage is available at WL. Lower CMOS inverter makes both the stacked NMOS on and at FS/125C, N1 and N2 are fast, and P1 is slow hence, the WL voltage is lowered to a large extent. Contrary to this, at SS/-40C, P1 is fast but N1 and N2 are slow therefore very less current flows through the stacked NMOS and maximum current flows through the fast PMOS which leads to very less WL underdrive at SS and SF corners thereby preventing WLUD at cell current and WM critical PVT corners. This scheme proves to be advantageous in improving the SNM without putting significant penalty on the cell current and WM at their respective critical PVTs.

IV. WRITE ASSIST CIRCUIT

WM of the cell degrades at lower operating voltages so there is a need to ensure the write-ability of the cell. Write assist scheme is used to meet the desired specifications for the WM. Since the read assist scheme (compensated WLUD) which we are using here degrades the WM of the cell hence write assist is deployed to improve the degraded WM and to qualify its desired cell sigma. There are several write assist schemes that have been proposed so far [2][9-14]. We implement Bitline Under-shoot using capacitive coupling approach as write assist. Test setup of Write assist circuit along with a Write Driver and 1000 SRAM cells in a column is shown in Fig. 2.

A capacitor is added on the bitline, which couples negative charge on the bitline. When *BL_Drive* voltage is suddenly changed from high to low, then capacitor couples negative charge on BLB thereby providing required Bitline under-shoot. Fig. 3. shows the timing diagram about the Capacitively Coupled scheme. In this work, a capacitance of 10fF is used

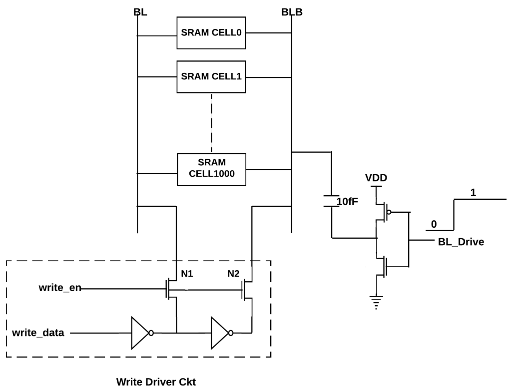

Fig. 2. Simulation setup showing Capacitive Coupling Circuit with SRAM cells and Write Driver

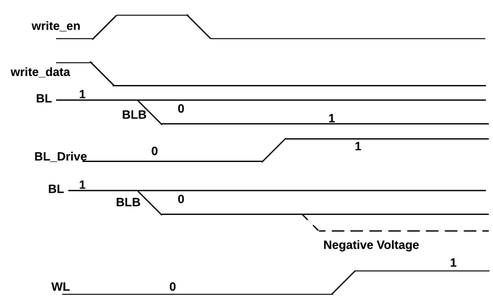

Fig. 3. Timing Diagram For Write Using Write Assist Circuit

to obtain the required bitline under-drive, thereby enhancing the WM to qualify required cell sigma.

By this assist technique write-ability and write time improves significantly providing negligible impact on cell stability, read speed and leakage[9]. Due to the additional capacitor there is a limited area overhead which is acceptable as the capacitor is small.

V. Simulation Setup And Results

A. Read Assist Circuit

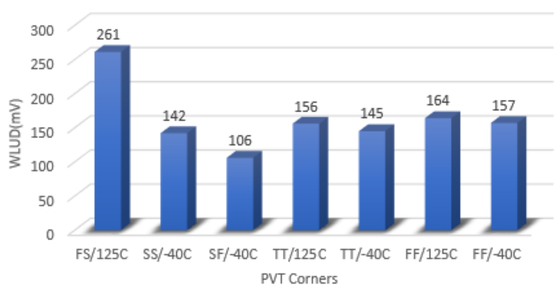

Fig. 4. WLUD achieved at various PVT corners at VDD=0.9V

TABLE III
SNM, Icell and WM For Different PVT @0.9V(After Read Assist)

PVT	SNM		WM		Icell
	(mV)	Cell sigma	(mV)	Cell sigma	(μA)
FS/125C	465	9.20	151	4.38	5.34
SS/-40C	424	7.79	131	4.03	3.67
SF/-40C	426	8.21	141	4.38	4.99
TT/125C	403	7.72	205	6.08	6.78
TT/-40C	425	7.78	150	4.57	5.57
FF/125C	394	7.58	210	6.21	8.89
FF/-40C	428	8.04	158	4.75	7.85

With a compensated WLUD approach implemented on the SRAM cell, WLUD achieved for FS/0.9V/125C is 261mV. SNM for this PVT is 465mV and it is 9.20 sigma qualified. Fig. 4. shows the obtained WLUD at various PVT corners. SNM, WM, and Icell along with Monte Carlo Simulation results for SNM and WM are depicted in Table III. These results very clearly states that the WLUD obtained for FS corner is maximum whereas for SS and SF corner, it is almost negligible. Hence, it could be depicted that there is an improvement in SNM with a minimal impact on the cell current, WM at their respective critical PVTs. Fig. 5. depicts comparison of SNM sigma qualification for different PVT corners before and after read assist.

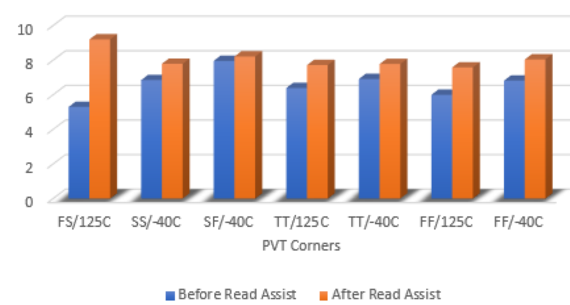

Fig. 5. SNM sigma at various PVT corners at VDD=0.9V

B. Write Assist Circuit

TABLE IV
WM And its Sigma Qualification For Different PVT @0.9V(After Read + Write Assist)

PVT	WM	
	(mV)	sigma
FS/125C	312	8.51
SS/-40C	323	9.96
SF/-40C	286	8.86
TT/125C	365	10.13
TT/-40C	296	9.03
FF/125C	366	10.2
FF/-40C	302	9.15

Without any write assist scheme, the WM of the SRAM cell is 148mV, and it is 4.59 sigma qualified for the worst

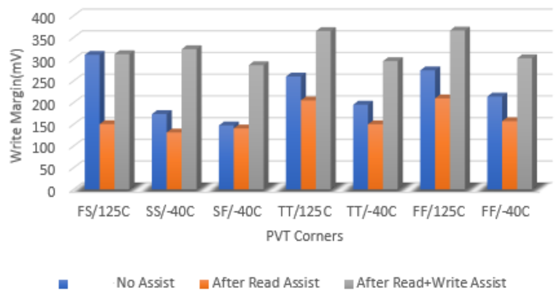

Fig. 6. WM achieved at various PVT corners at VDD=0.9V

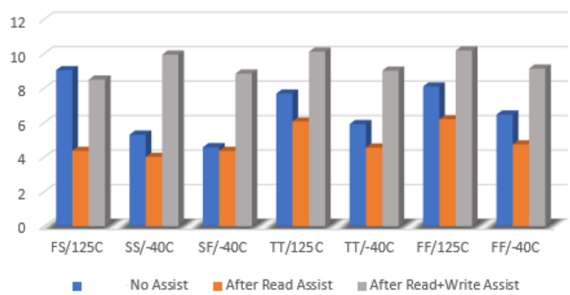

Fig. 7. WM sigma at PVT corners at VDD=0.9V

PVT corner i.e., SF/0.9V/-40C. Using Capacitive Coupling approach as write assist, bitline under-shoot obtained is 166mV for SF/0.9V/- 40C. WM for this PVT is 286mV, and it is 8.86 sigma qualified. From Table IV, WM and its Monte Carlo Simulation results can be observed. Fig. 6. and Fig. 7. shows comparison of WM values and their corresponding sigma when no assist scheme is applied, only read assist applied and after applying both read and write assist scheme. Hence observing the results, it can be stated that the Write assist scheme improved the WM of the cell to qualify desired cell sigma.

VI. Conclusion

In this work, we designed a high density 6T SRAM cell whose Vmin without any assist scheme is 1.2V. We have lowered the Vmin from 1.2V to 0.9V using read and write assist circuits. Compensated WLUD circuit is designed, simulated as read assist and deploying it we improved SNM at FS/0.9V/125C. This read assist circuit provides 261mV of WL underdrive at FS/125C which improves the SNM from 302mV to 465mV at this PVT. The SNM cell sigma improves from 5.30 sigma to 9.2 sigma ensuring high yield for a capacity of 64 Mb SRAM. This read assist scheme improves the SNM without significantly degrading the WM and speed of the cell at their respective critical PVT.

Write assist circuit was designed and simulated for a range of PVT corners. This scheme improved the WM of the cell to qualify desired cell-sigma at 0.9V. Negative Bitline scheme is implemented with the help of a Capacitive Coupling. BL underdrive of 166mV is obtained which ensures WM improvement from 148mV to 286mV at SF/-40C with WM being 8.86 sigma qualified.

We demonstrate 300mV voltage lowering by using read and write assist schemes without significant area penalty thereby enabling upto 45% reduction in the overall dynamic power of the design.

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